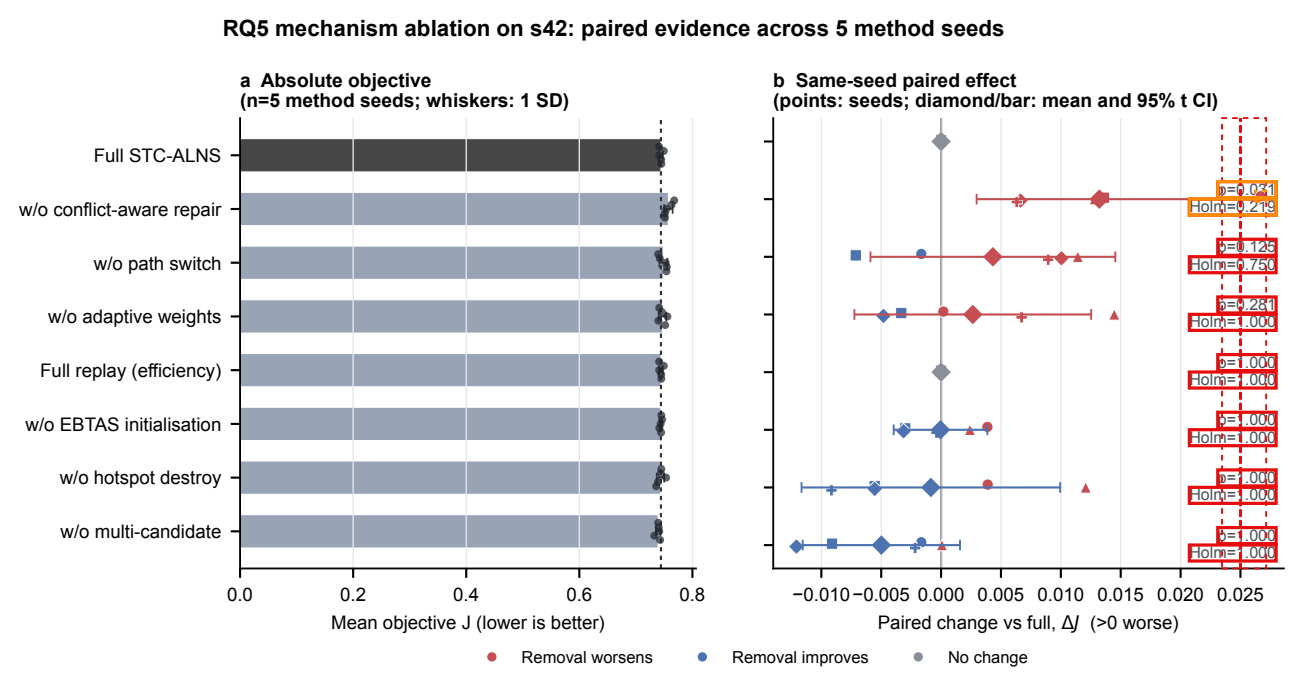

RQ5 mechanism ablation on s42: paired evidence across 5 method seeds
a Absolute objective
(n=5 method seeds; whiskers: 1 SD)
Full STC-ALNS
w/o conflict-aware repair
w/o path switch
w/o adaptive weights
Full replay (efficiency)
w/o EBTAS initialisation
w/o hotspot destroy
w/o multi-candidate
0.0
0.2
0.4
0.6
0.8
Mean objective J (lower is better)
b Same-seed paired effect
(points: seeds; diamond/bar: mean and 95% t CI)
p=0.031
Holm=0.219
p=0.125
Holm=0.750
p=0.281
Holm=1.000
p=1.000
Holm=1.000
p=1.000
Holm=1.000
p=1.000
Holm=1.000
p=1.000
Holm=1.000
−0.010
−0.005
0.000
0.005
0.010
0.015
0.020
0.025
Paired change vs full, ΔJ (>0 worse)
Removal worsens
Removal improves
No change